CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharmacy Examination

University Theory Examination November – December 2017

PH303 Medicinal Chemistry-I

Date: 25/11/2017, Saturday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. **Figures to right indicate marks.**
2. **Section wise separate answer book is to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any TWO of the following:

A Write the comments on :(ANY FOUR) **10**

1. Pyridine is more basic than pyrrole.
2. Electrophilic substitution takes place at 2nd and 5th position in pyrrole.
3. Impurities of thiophene in benzene can be removed by cold sulphuric acid.
4. Imidazole is more basic than pyrazole.
5. Proton Pump Inhibitors acts as Pro-drugs.

B 1. Discuss the relationship between biological action and physicochemical properties for ANY ONE of the following with suitable example: **05**

 1. Partition Coefficient
 2. Hydrogen bonding

2. Explain any two Phase I metabolic reactions along with suitable examples. **05**

C 1. Draw the structure of: **05**

(1) Furan (2) Indol (3) Isoquinoline (4) Triazole, (5) Pyran

2. Give synthesis of Indole formation along with it's reaction mechanism. **05**

Q 2 Attempt any TWO of the following:

A Write classification, SAR and chemical mechanism of action of Proton Pump Inhibitors. Give scheme of synthesis of Pentoprazole. **10**

B 1. Give Proton Transfer (Hantzsch) synthesis for Pyridine with reaction mechanism. **05**

2. Write Skraup's synthesis with reaction mechanism. **05**

C 1. Give any TWO methods of preparation of the following: **05**

(1) Pyrrole (2) Furan (3) Thiophen (4) Imidazole

2. Give SAR of H_2-Antagonists. **05**

SECTION- II

Q 3		**Attempt any <u>TWO</u> of the following:**	
A	1.	With the help of a neat labeled diagram discuss the mechanism of action of Cholinesterase Inhibitors.	**07**
	2.	Classify Antacids.	**03**
B		Discuss the classification of Adrenergic agonist with suitable examples and discuss the SAR of Adrenergic agonist (Phenyethanolamine agonists).	**10**
C	1.	Classify and write the SAR of Cholinergic agonist.	**07**
	2.	Give synthesis of Propranolol.	**03**
Q 4		**Attempt any <u>TWO</u> of the following:**	
A		What are Pro-drugs? Classify Pro-drugs and Give suitable examples of each type of pro-drug.	**10**
B		Classify Antitussives and Expectorants with the help of chemical structures. Give the mechanism of action of Expactorants.	**10**
C	1.	Write a short note of Anti-asthmatic agents.	**05**
	2.	Give Classification of Laxatives and Antidiarrheals.	**05**
